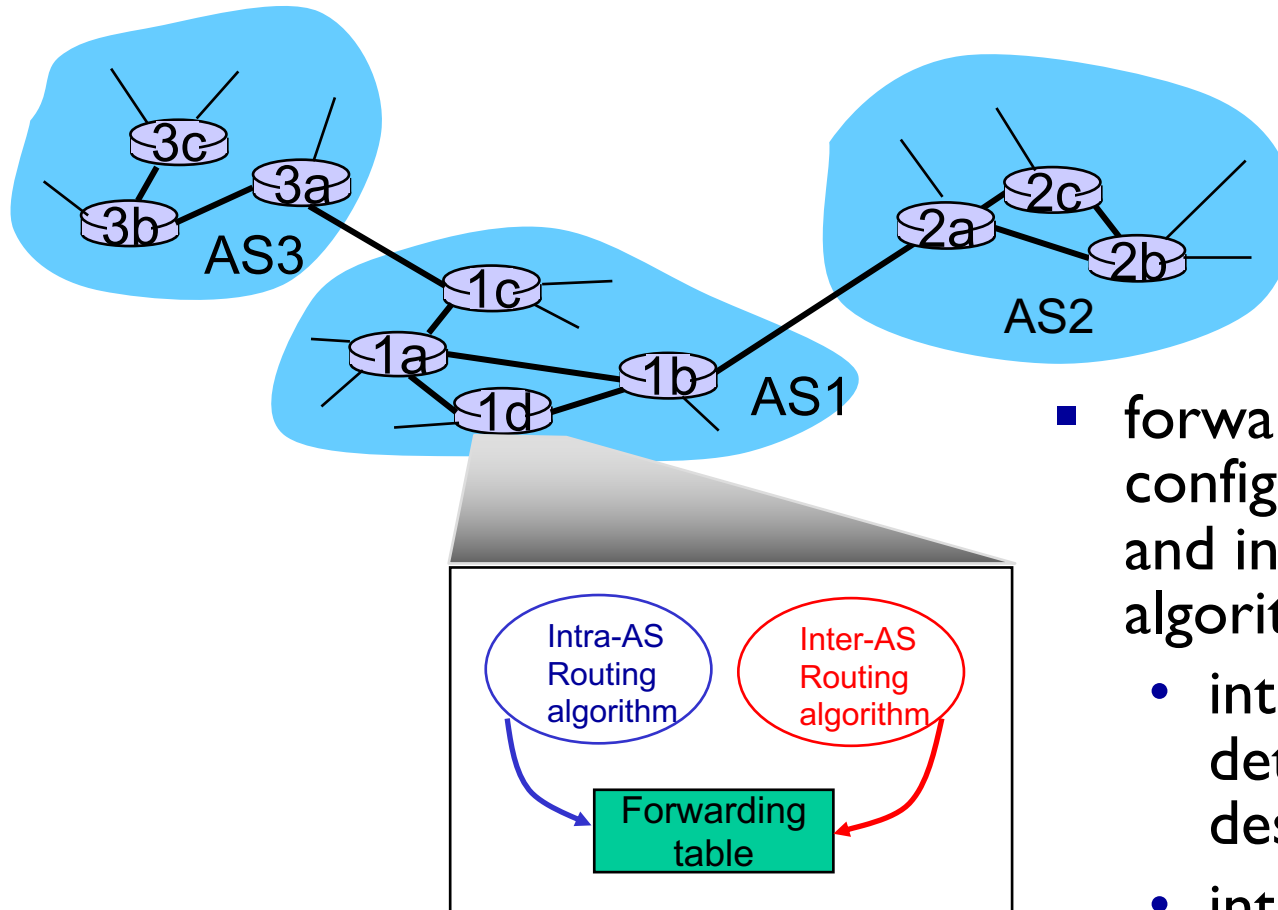


Intra-AS Routing

- also known as *interior gateway protocols (IGP)*
- most common intra-AS routing protocols:
 - RIP: Routing Information Protocol
 - OSPF: Open Shortest Path First (IS-IS protocol essentially same as OSPF)
 - IGRP: Interior Gateway Routing Protocol (Cisco proprietary for decades, until 2016)

Interconnected ASes



- forwarding table configured by both intra- and inter-AS routing algorithm
 - intra-AS routing determine entries for destinations within AS
 - inter-AS & intra-AS determine entries for external destinations

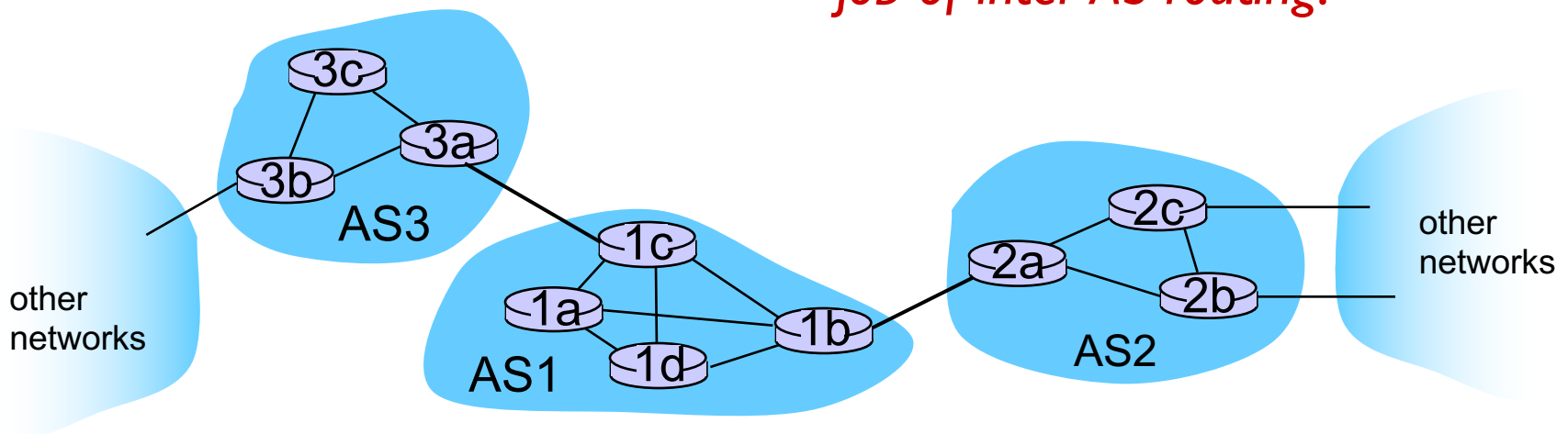
Inter-AS tasks

- suppose router in AS1 receives datagram destined outside of AS1:
 - router should forward packet to gateway router, but which one?

AS1 must:

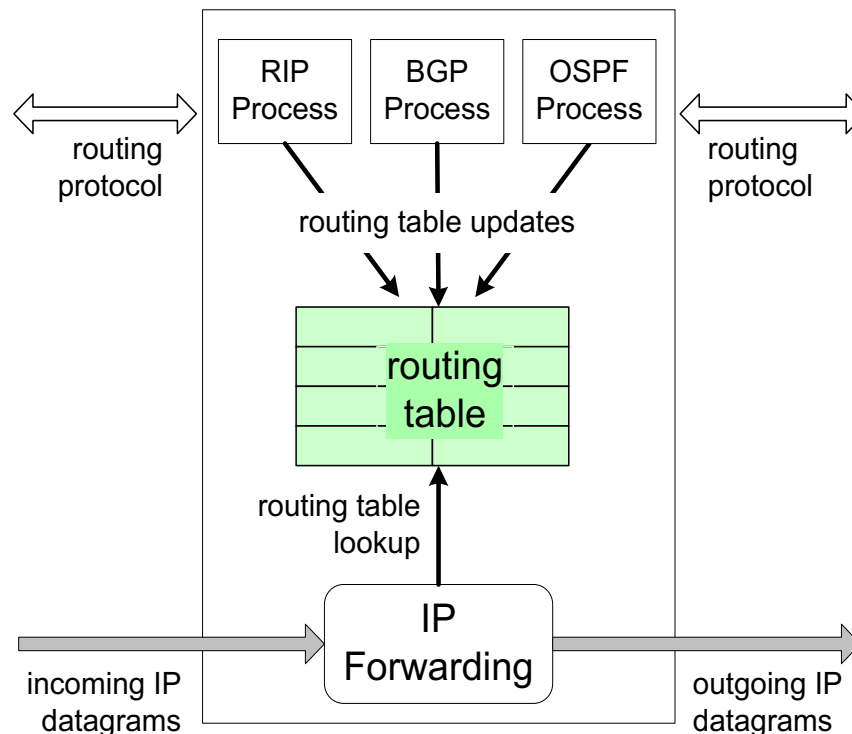
1. learn which destinations are reachable through AS2, and which through AS3
2. propagate this reachability info to all routers in AS1

job of inter-AS routing!



Multiple Routing Protocols

- multiple routing protocols can run on the same router
- if a router is an exterior gateway router then usually one IGP and one EGP protocol will be in operation
- each routing protocol updates the routing table accordingly



Chapter 5: outline

5.1 introduction

5.2 routing protocols

- link state
- distance vector

5.3 intra-AS routing in the Internet:

- RIP
- OSPF

5.4 inter-AS routing in the Internet: BGP

Routing protocols

Routing protocol goal: determine “good” paths (equivalently, routes), from sending host to receiving host, through network of routers

- path: sequence of routers packets will traverse in going from given initial source host to given final destination host
- “good”: “least cost”, “fastest”, “least congested”
- routing: a “top-10” networking challenge!

Components of a Routing Algorithm

- a procedure for **sending and receiving** reachability information about a network to other routers
- a procedure for **calculating optimal** routes
 - Routes are calculated using a shortest path algorithm (least “**cost**”)
- a procedure for **reacting to and advertising** topology changes

Routing algorithm classification

Q: global or local information?

global:

- all routers have complete topology, link cost info
- “link state” algorithms

local:

- router knows physically-connected neighbors, link costs to neighbors
- iterative process of computation, exchange of info with neighbors
- “distance vector” algorithms

Q: quasi-static or dynamic?

quasi-static:

- routes change slowly over time

dynamic:

- routes change more quickly
 - periodic update
 - in response to link cost changes

Two Shortest Path IGP Routing Algorithms

Distance Vector Routing

- each node knows the distance (cost) to its directly connected neighbors
- a node periodically sends a list of routing updates to its neighbors
- if all nodes update their distances to destinations using neighbor information, the routing tables eventually converge

Link State Routing

- each node knows the distance (cost) to its directly connected neighbors
- the distance information is flooded to all nodes in the network
- each node calculates the routing tables independently using a network map (topology) created by the node using the global information it received

Chapter 5: outline

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Routing Information Protocol (RIP)

- “open”: publicly available
- uses distance vector algorithm
 - routing table dissemination
 - only link cost to neighbors at each node
 - route computation using Bellman Ford’s algorithm
- router advertises its routing table to all its neighbors
 - carried in RIP messages over UDP
 - routing table entries give cost from that node (source) to all other nodes (destinations) in AS

RIP: Basic principles

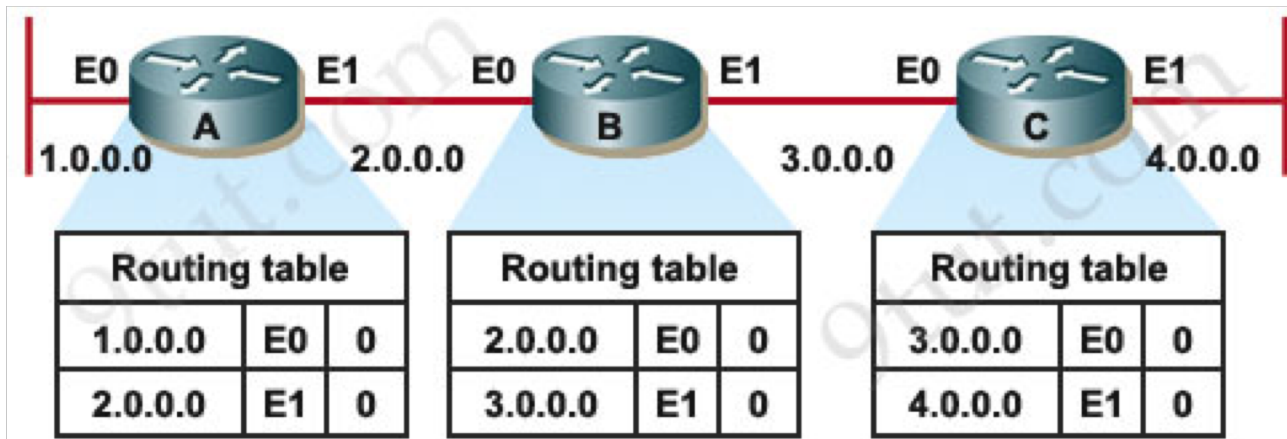
- periodically, every router sends to its neighbors a complete list of its routes to all destinations within an AS
- list contains pairs of: **destination**, **distance**
- receiver replaces/updates entries in its routing table if routing through a **neighbor** costs less than the current route in its table

Rip Example

assume:

- link cost is 1 on all hops
- all updates occur simultaneously

initially each router only knows its directly connected interfaces -> cost = 0



After First Update

